A Unified Approach to Optimization via Generalized Moment Problems and (p, q)-Mean Polynomials

1. Introduction

Optimization problems, seeking to minimize (or maximize) a function subject to constraints, are fundamental to numerous fields. While Lasserre's SDP hierarchy has revolutionized polynomial optimization, its application to broader classes of functions remains limited. This proposal outlines a novel approach to extend Lasserre's methodology by unifying recent advances in generalized moment problems, (p, q)-mean polynomials, and techniques from differential algebra. This unified framework will enable the optimization of functions with nice algebraic representations, including those arising from algebraic differential equations.

2. Background and Motivation

2.1 Optimization and its Challenges

The core optimization problem is:

$$(*) \quad \begin{cases} \text{minimize} & f(x) \\ \text{subject to :} & \\ & g_i(x) \geq 0, \quad i = 1, \ldots, k. \end{cases}$$

Solving $(*)$ is NP-hard, even for polynomials. Lasserre's SDP hierarchy offers a powerful solution for polynomial optimization by exploiting Positivstellensatz results and the truncated moment problem. However, generalizing this to functions beyond polynomials, such as exponentials or trigonometric functions with interconnected variables, presents a significant hurdle.

2.2 Moment Problems and Algebraic Representations

Recent work has extended the theory of moment problems from polynomials to semigroup algebras and commutative unital algebras [GK01, GMW, GKM01, CGIK]. This provides a foundation for applying moment-based optimization to a broader class of functions.

2.3 The Power of (p, q)-mean Polynomials

The generalized weighted p-mean and the resulting (p, q)-mean polynomials are key to establishing Positivstellensatz results. These polynomials offer a powerful tool for representing positive polynomials and deriving inequalities crucial for optimization.

3. Research Objectives

This proposal aims to achieve the following:

- **Objective 1:** Develop a comprehensive theory of the truncated moment problem for semigroup algebras, building upon the results in [CGIK]. This will enable the construction of SDP hierarchies for functions within these algebras.
- **Objective 2:** Develop robust methods to incorporate dependencies between variables in the optimization framework. This is vital for handling functions like x and e^x, where treating variables as independent leads to incorrect solutions.
- **Objective 3:** Integrate KKT conditions and Lagrangian analysis into the SDP hierarchy. This will incorporate derivative information and optimality conditions, enhancing the method's applicability to functions with algebraic representations.

4. Proposed Methodology

4.1 Extending the Truncated Moment Problem

We will leverage the results in [CGIK] to develop a robust theory of the truncated moment problem for semigroup algebras. This involves:

- Defining appropriate moment matrices and analyzing their properties.
- Establishing conditions for the existence and uniqueness of representing measures.
- Developing efficient algorithms for solving the truncated moment problem in this setting.

4.2 Incorporating Dependencies

We will investigate these approaches to capture variable dependencies:

- **Differential Algebra:** Represent and manipulate functions with interconnected variables using the tools of differential algebra.
- **Radon-Nikodym Theorem:** Explore the relationship between representing measures and their derivatives using the Radon-Nikodym theorem.
- **KKT Conditions:** Integrate KKT conditions into the SDP hierarchy, enforcing relationships between variables and their derivatives at optimal solutions.

Lagrangian

The Lagrangian of the above problem is defined as

$$L(f, \mathbf{g}, \lambda) = f - \sum_{i=1}^{k} \lambda_i g_i,$$

where $\lambda \in \mathbb{R}^n_{\geq 0}$. Lagrangian incorporates both the objective function and the constraints into a single expression, making it easier to analyze the problem mathematically. Let x^* be an optimal solution to the primal problem (the original minimization problem). Let λ^* be an optimal solution to the dual problem (the problem of maximizing the Lagrangian for λ, subject to $\lambda \geq 0$). Then, for any feasible point x (i.e., a point that satisfies all the constraints $g_i(x) \geq 0$) and any $\lambda \in \mathbb{R}^n_{\geq 0}$, the following inequality holds:

$$L(f, \mathbf{g}, \lambda)(x^*) \leq f(x^*) \leq L(f, \mathbf{g}, \lambda^*)(x),$$

where

- x^* is an optimal solution to the primal problem (the original minimization problem)
- λ^* is an optimal solution to the dual problem (the problem of maximizing the Lagrangian for λ, subject to $\lambda \geq 0$)
- x is any feasible point (satisfies all constraints $g_i(x) \geq 0$)
- λ is any vector of non-negative Lagrange multipliers ($\lambda \in \mathbb{R}^n_{\geq 0}$).

The left-hand side inequality $L(f, \mathbf{g}, \lambda)(x^*) \leq f(x^*)$ states that the value of the Lagrangian at the optimal primal solution x^* and any nonnegative Lagrange multipliers λ provides a **lower bound** on the optimal value of the primal objective function $f(x^*)$. The right-hand side inequality $f(x^*) \leq L(f, \mathbf{g}, \lambda^*)(x)$ states that the value of the Lagrangian at any feasible point x and the optimal dual variables λ^* provides an **upper bound** on the optimal value of the primal objective function $f(x^*)$.

In real algebraic geometry, the Lagrangian has been used to estimate the lower bound for the optimal $(*)$ value. Articles [GM01, GM02, GLM, DIdW01, DIdW02] provide examples of the effort made in this line.

KKT (Karush-Kuhn-Tucker conditions)

KKT conditions are a set of necessary conditions for optimal solutions to a constrained optimization problem. They are named after Harold W. Kuhn and Albert W. Tucker, who independently discovered them in 1951.

The KKT conditions state that, at an optimal solution, the following conditions must hold:

- The gradient of the objective function must be equal to a linear combination of the gradients of the constraint functions multiplied by the Lagrange Multipliers.
- The Lagrange multipliers must be nonnegative.
- The constraint functions must be satisfied at the optimal solution.

The KKT conditions can be used to find the optimal solution to a constrained optimization problem by solving a system of nonlinear equations. However, it is essential to note that the KKT conditions are only necessary for optimality and not sufficient. This means that a solution that satisfies the KKT conditions may not be the global optimal solution while keeping the framework within the context of semigroup-algebras.

KKT Conditions The KKT conditions provide the conditions for a solution to be optimal in a nonlinear optimization problem with inequality constraints. They are:

1. **Stationarity:** $\nabla f(x^*) - \sum \lambda_i \nabla g_i(x^*) = 0$,
2. **Primal feasibility:** $g_i(x^*) \geq 0$ for $i = 1, \ldots, k$,
3. **Dual feasibility:** $\lambda_i \geq 0$ for $i = 1, \ldots, k$,
4. **Complementary slackness:** $\lambda_i g_i(x^*) = 0$ for $i = 1, \ldots, k$. Where:

- x^* is the optimal solution,
- $\nabla f(x^*)$ is the gradient of the objective function at x^*,
- $\nabla g_i(x^*)$ is the gradient of the i-th constraint function at x^*,
- λ_i are the KKT (Lagrange multipliers) associated with each constraint.

Given a consistent set of derivatives, including the KKT condition in the SDP problem, relations amongst dependent and independent symbols in differentials are incorporated.

> **Note:** Investigating the conditions under which using SDP hierarchies together with KKT conditions results in the generalization of Lasserre's method to a more general class of functions.

4.3 Leveraging (p, q)-mean Polynomials and Positivstellensatz

We will utilize (p, q)-mean polynomials to:

- Construct Positivstellensatz certificates for functions in semigroup algebras, leading to stronger and more general representations.
- Derive tighter bounds for optimization problems by exploiting the inequalities arising from these polynomials.
- Refine existing Positivstellensatz results (e.g., [DIdW02, DM]) by incorporating the framework of semigroup algebras and (p, q)-mean polynomials.

$(p, q)-$mean polynomials

Lets $p \in \mathbb{R}$ and $\mathbf{x} = (x_1, \ldots, x_), w \in \mathbb{R}_+^n$. The *generalised weighted p-mean* is defined by

$$M_p(\mathbf{x}, w) = \left(\frac{\sum_{i=1}^n w_i x_i^p}{\sum_{i=1}^n w_i} \right)^{1/p}.$$

Proposition 1

- $\lim_{p \to -\infty} M_p(\mathbf{x}, w) = \min(x_1, \ldots, x_n)$.
- $\lim_{p \to \infty} M_p(\mathbf{x}, w) = \max(x_1, \ldots, x_n)$.
- $\lim_{p \to 0} M_p(\mathbf{x}, w) = (\prod_{i=1}^n x_i^{w_i})^{1/\sum_{i=1}^n w_i}$.
- If $p < q$, then $M_p(\mathbf{x}, w) \leq M_q(\mathbf{x}, w)$. Equality happens when $x_1 = x_2 = \cdots = x_n$.

Theorem 2 (Fidalgo-Kovacec/Ghasemi-Marshall) For a form $f = \sum_{i=1}^n \beta_i X_i^{2d} - \mu \underline{X}^\alpha$ with $\alpha \in \mathbb{N}^n$, $|\alpha| = 2d$, $\beta_i \geq 0$ for $i = 1, \ldots, n$, the followings are equivalent: - f is PSD. - $\mu^{2d} \prod_{i=1}^n \alpha_i^{\alpha_i} \leq (2d)^2 d \prod_{i=1}^n \beta_i^{\alpha_i}$.

Proof. Apply Proposition 1 for $M_{2d}(\underline{X}, \alpha) \geq M_0(\underline{X}, \alpha)$ to $\underline{X} = \left((\frac{\beta_1}{\alpha_1})^{1/2d} X_1, \ldots, (\frac{\beta_n}{\alpha_n})^{1/2d} X_n \right)$. $\square$

Theorem 3 (Dressler-Iliman-de Wolff) Let $\mathbf{0} = \alpha(0), \alpha(1), \ldots, \alpha(r) \in \mathbb{N}^n$ be linearly independent vertices of a polytope and $\beta \in \mathrm{conv}(\alpha(0), \ldots, \alpha(\mathrm{r}))^\circ \cap \mathbb{N}^n$ such that $\beta = \sum_{j=1}^r \lambda_j \alpha(r)$ with $\lambda_j > 0$ and $\sum_{j=0}^r \lambda_j = 1$, then for $f = \sum_{j=0}^r f_{\alpha(j)} \underline{X}^{\alpha(j)} + f_\beta \underline{X}^\beta$, the followings are equivalent: - f is PSD. - $|f_\beta| \leq \prod_{j=0}^r \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j}$.

Proof. Apply Proposition 1 for $M_1(\underline{X}, \lambda) \geq M_0(\underline{X}, \lambda)$ to $\underline{X} = (\frac{f_{\alpha(0)}}{\lambda_0} \underline{X}^{\alpha(0)}, \ldots, \frac{f_{\alpha(r)}}{\lambda_r} \underline{X}^{\alpha(r)})$. $\square$

Positivstellensatz

It is known that in the polynomial case, for finitely generated preorderings, being Archimedean and having the compact non-negativity set are equivalent (Wormann's theorem). This results in denominator-free Positivstellensatz like Schmudgen's. In non-polynomial case, lake of such theorems results in complex forms of statements. It is possible to replace the polynomial algebra with a

finitely generated commutative algebra and derive variations of Wormann's and Schmudgen's theorems. This will simplify and strengthen some existing Positivstellensatz (e.g., [DIdW02, DM, ...]).

Definition: Let $0 \leq p < q$, and $g_1, \ldots, g_s \in \mathbb{R}[\underline{X}]$. We denote by T_S the preprime generated by g_i's and $M_S^{p,q}$ the T_S-module generated by all instances of $M_q(\underline{X}, w)^{p+q} - M_p(\underline{X}, w)^{p+q}$ that are polynomials. We call these polynomials a (p, q)-mean polynomial.

Theorem (General Mean Positivstellensatz) If K_S is compact and $f \in \mathbb{R}[\underline{X}]$ is strictly positive on K_S, then there are (p, q)-mean polynomials $m_1, \ldots, m_k$ and $h_1, \ldots, h_k \in T_S$ such that $f = \sum_{i=1}^{k} m_i h_i$.

Proof: By Wormann's trick $M_S^{p,q}$ is Archimedean and hence the representation theorem applies. $\square$

5. Research Plan Break Down

- Develop the theory and algorithms for the truncated moment problem in semigroup algebras.
- Explore the application of differential algebra to represent and manipulate functions with dependencies.
- Conduct preliminary investigations into integrating KKT conditions into the SDP hierarchy.
- Implement and test algorithms for solving the truncated moment problem in the context of semigroup algebras.
- Investigate the use of the Radon-Nikodym theorem to capture dependencies between variables.
- Develop and implement algorithms that combine the SDP hierarchy with KKT conditions.
- Apply the unified framework to concrete optimization problems from various domains, including those involving solutions to algebraic differential equations.
- Analyze the performance of the algorithms and refine the approach based on experimental results.
- Disseminate research findings through publications in top-tier journals and presentations at international conferences.

6. Expected Outcomes and Impact

This research is expected to:

- **Significantly extend the scope of Lasserre's SDP hierarchy** to encompass a much wider class of functions.
- **Develop new theoretical and computational tools** for optimization in semigroup algebras, unifying moment problems, (p, q)-mean polynomials,

and differential algebra.

- **Advance the understanding of Positivstellensatz** and its applications in optimization.
- **Provide practical solutions to challenging optimization problems** in diverse fields, including control theory, robotics, engineering, and machine learning.

By developing this unified framework, this research aims to make a significant contribution to both the theory and practice of optimization, enabling the solution of problems previously considered intractable.

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